

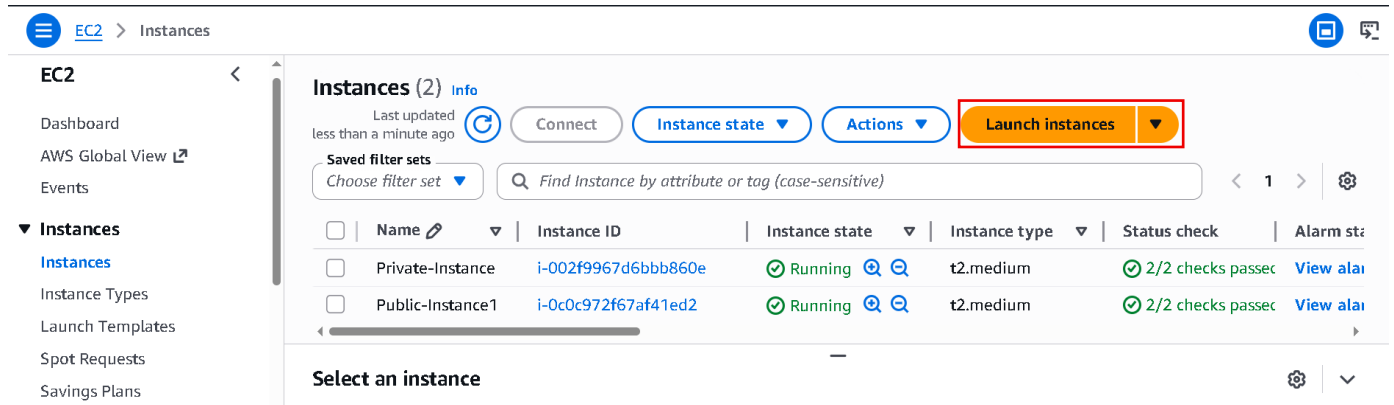
Linux EC2 Instance

Summary: -

This document explains how to create a Linux EC2 instance in AWS, configure networking and security groups, create a key pair, and connect to the instance using SSH from the command line with the .pem key file and public IP address.

Let's Create: -

- Search EC2 in Aws Console and Open it.
- Now Let's create windows EC2 instance.



EC2 > Instances

Instances (2) Info

Last updated less than a minute ago

Connect Instance state Actions **Launch instances**

Saved filter sets Choose filter set

Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm sta
☐	Private-Instance	i-002f9967d6bbb860e	Running	t2.medium	2/2 checks passed	View alai
☐	Public-Instance1	i-0c0c972f67af41ed2	Running	t2.medium	2/2 checks passed	View alai

Select an instance

- Now opens EC2 in new tab.
- And click Launch instance.

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

MyFirstLinux

Add additional tags

- Enter instance name.
- Scroll down.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

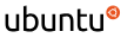
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

Quick Start


Amazon Linux


macOS


Ubuntu


Windows


Red Hat


SUSE


[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

- Click Amazon Linux.
- scroll down.

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Linux base pricing: 0.0104 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

- now we are just going to select this t3.micro
- In this we are going to get two virtual CPU and one GB Ram.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Select

 [Create new key pair](#)

- Click Create new key pair.

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

demo

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

Cancel

Create key pair

- Enter key name.
- Key pair type as RSA
- And .pem file
- And last click Create key pair
- Now your keypair is downloaded and save it.
- Now scroll down.

▼ Network settings

Info

Edit

Network

Info

vpc-072c4acd761c3b942

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Additional charges apply when outside of free tier allowance

- Now click Edit.

▼ Network settings [Info](#)

VPC - *required* | [Info](#)

vpc-072c4acd761c3b942
172.31.0.0/16

(default) ▼



Subnet | [Info](#)

subnet-04ffa8e5e9c79ad3d
VPC: vpc-072c4acd761c3b942 Owner: 463646775279
Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone
IP addresses available: 4089 CIDR: 172.31.32.0/20

[Create new subnet](#)

Auto-assign public IP | [Info](#)

Enable ▼

Additional charges apply when outside of [free tier allowance](#)

- Now Default VPC not change it.
- Select Subnet ab-south-1a because in Mumbai region three availability zones 1a, 1b, 1c if we not choose then aws decide in which availability zones they want to create our EC2 instance.

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - *required*

launch-wizard-80

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . _ - / 0 #, @ [] + = & ; [] ! \$ *

Description - *required* | [Info](#)

launch-wizard-80 created 2026-05-19T08:26:27.229Z

- Now here create a security group or you select your created security group also.
- Aws always create a custom security group if you want to change name of security group you can change it.

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

[Remove](#)

Type | [Info](#)

ssh ▼

Protocol | [Info](#)

TCP

Port range | [Info](#)

22

Source type | [Info](#)

Anywhere ▼

Source | [Info](#)

Add CIDR, prefix list or security group

0.0.0.0/0

Description - *optional* | [Info](#)

e.g. SSH for admin desktop

- You can add extra permissions in your security group.
- By default for Linux ssh permissions is added.

- Now scroll down.

▼ Configure storage [Info](#)

Advanced

1x GiB ▼ Root volume, 3000 IOPS, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

[Add new volume](#)

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

- Suppose if you are going to buy a laptop, they will ask you right that what is the storage Requirement like One TB or 512 GB etc.
- Same way here.
- It will provide you 8 GB volume.
- Now this volume is actually root volume.
- It means it will install operating system inside this volume.

ami-09ed39e30153c3bf9

[Virtual server type \(instance type\)](#)

t3.micro

[Cancel](#)

[Launch instance](#)

[Preview code](#)

- Now click Launch instance.

Instances (1/3) [Info](#)

Last updated
5 minutes ago

[Connect](#)

[Instance state](#) ▼

[Actions](#) ▼

[Launch instances](#) ▼

Saved filter sets

[Choose filter set](#) ▼

Find Instance by attribute or tag (case-sensitive)

< 1 >

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4
☒	MyFirstLinux	i-0c097d3773e5222a4	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-233-196-178

i-0c097d3773e5222a4 (MyFirstLinux)

▼

[Details](#)

[Status and alarms](#)

[Monitoring](#)

[Security](#)

[Networking](#)

[Storage](#)

[Tags](#)

▼ Instance summary [Info](#)

Instance ID

i-0c097d3773e5222a4

Public IPv4 address

13.233.196.178 | [open address](#)

Private IPv4 addresses

172.31.37.97

- Now it's ready to use.
- Now we can access with command Line because it has only ssh inbound rule.

- ```
C:\ ec2-user@ip-192-168-0-45::~~
Microsoft Windows [Version 10.0.19045.7184]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akash Kumar>cd Downloads

C:\Users\Akash Kumar\Downloads>ssh -i demo.pem ec2-user@13.207.2.165
The authenticity of host '13.207.2.165 (13.207.2.165)' can't be established.
ED25519 key fingerprint is SHA256:u5ZyuJ65FhjTP9UYG5J45trh5J/NX2FQilFNogIIE4s.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.207.2.165' (ED25519) to the list of known hosts.

, #_
~_ #####_ Amazon Linux 2023
~~ \#####\
~~ \|###|
~~ \|#/ _____ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~'->
~~~~ /  
~-.- _/  
_/m/'
```
- [ec2-user@ip-192-168-0-45 ~]\$ █